PROFESSIONAL PORTFOLIO

Rawad Abi Naim

AI Software Engineer | Full Stack Developer | AI and Data Systems | DevOps | Cybersecurity

Ontario, Canada | rawadnm15@gmail.com | LinkedIn | GitHub

Professional Profile

I build software with a practical objective: turn technical capability into measurable operational value. My work combines full stack development, artificial intelligence, data analysis, cloud technologies, security engineering, and visual communication. I am particularly interested in systems where software does more than store information and instead helps people decide what to do next.

My project experience includes AI based lead intelligence, predictive analytics, security assessment, cloud connected applications, autonomous systems, and business software. Across these projects, I focus on the complete path from data and architecture to user experience, deployment, measurement, and business impact.

The quantitative results in this portfolio are reported project or pilot use case results from my source materials. They describe observed or estimated outcomes in those contexts and are not presented as universal guarantees.

Core Value I Bring

- Translate business problems into practical software and AI workflows.
- Build data driven applications that prioritize decisions instead of simply presenting information.
- Combine machine learning with dashboards, APIs, cloud services, and secure application design.
- Measure improvements through cost reduction, time saved, conversion efficiency, engagement, availability, and risk reduction.
- Bridge technical and visual requirements so complex systems remain understandable to business users.

Selected AI and Business Intelligence Projects

Autonomous Character Robotics and Embodied AI Platform

The Autonomous Character Robotics project was developed as a fully offline, multi-modal embodied AI system built around an articulated five-degree-of-freedom (5-DOF) lamp robot. Rather than coupling raw generative models directly to motor hardware, the system establishes a production-grade boundary between laptop-side cognition and an authoritative body runtime connected over a custom TCP socket network transport. The result is a goal-directed, privacy-first character workflow capable of live face engagement, offline speech recognition, scene memory reasoning, physical trajectory execution, and multi-modal sensory expression.

• Engineered an embodied AI architecture separating high-level multimodal cognition from authoritative physical execution across a dedicated TCP socket boundary.
• Implemented strict protocol and safety controls utilizing validated JSON schemas (BodyCommand and BodyResult), sequence tracking, monotonic intent clamping, and whole-command

timeout enforcement.
• Integrated a 100% offline local AI stack combining OpenCV for face tracking, Vosk for offline speech-to-text, Moondream VLM for visual scene understanding, and Qwen2 (0.5B) for structured intent planning on standard CPU hardware.
• Built a multi-modal expression engine synchronizing smooth PyBullet joint trajectory interpolation, dynamic RGB lamp shade lighting cues, algorithmic PCM sound effects and musical melodies, and local text-to-speech.
• Developed an autonomous observe–act–reobserve loop that extracts target objects from spoken user goals, inspects live camera visual evidence, initiates dynamic scene search sweeps when items are absent, and verifies physical task completion against joint settling thresholds (<0.15 rad).
• Designed automated telemetry and testing pipelines featuring 21 comprehensive unit and integration tests across protocol, transport, receiver, controller, and runtime verification, with persistent JSONL telemetry logging.
• Targeted lightweight, zero-cloud deployment optimized for clean Ubuntu LTS environments requiring no discrete GPU or external cloud dependencies.

Engineering value: The critical architectural contribution is enforcing an authoritative control boundary between cognitive reasoning and physical execution, ensuring that vision-language models can guide expressive autonomous behavior without compromising hardware safety, timing predictability, or data privacy.

Smart Leads AI and Business Intelligence Platform

The Smart Leads AI project was developed as a decision support platform for retailers and sales teams. Instead of treating every prospect equally, the system ingests customer and engagement information, normalizes the data, and applies predictive scoring to identify prospects with stronger purchase potential. The result is a workflow in which marketing teams can move from broad prospect review toward ranked, evidence based prioritization.

- Built an AI driven lead intelligence workflow ranking prospects using behavioral engagement, conversion likelihood, and marketing value.
- Developed an interactive dashboard for search, filtering, lead management, score analysis, and business growth insights.
- Integrated lead scoring, marketing recommendations, social analytics, and approval workflows into one decision support environment.
- Reported approximately 38 percent lower marketing costs in project use cases through more selective targeting.
- Reported a 31 percent improvement in qualified lead conversion efficiency through AI based prioritization.
- Reported a 24 percent improvement in campaign response rates through improved customer segment identification.
- Reported approximately CAD 10,000 in additional average monthly sales across pilot use cases.
- Reduced high value customer identification time by more than 70 percent compared with manual and spreadsheet based review.

Business value: The important contribution is not the score itself. The value comes from reducing attention spent on low probability prospects and directing limited marketing resources toward opportunities with stronger predicted value.

Retail Intelligence and Lead Prediction Engine

This platform transforms raw social marketing and customer signals into real time conversion probability scores and business intelligence views. It was evaluated with two clothing retail businesses during a pilot implementation.

- Designed a machine learning pipeline combining multiple customer engagement signals into real time conversion probability scores.
- Built dashboards for lead ranking, customer segmentation, filtering, campaign performance, and visualization.
- Reported a 42 percent reduction in customer acquisition and advertising costs through predictive targeting and campaign recommendations.
- Reported a 35 percent improvement in qualified lead conversion rates by prioritizing higher probability customers.
- Reported a 28 percent increase in campaign engagement through AI assisted segmentation and personalized recommendations.
- Reported approximately CAD 12,000 in combined monthly sales improvement across participating businesses.
- Reduced manual lead qualification time by approximately 75 percent.
- Designed scalable web architecture intended to support thousands of predictions with low latency and future model improvements.

Business value: Predictive analytics connects marketing activity to operational decisions by converting raw engagement signals into a ranked and actionable view.

Software Engineering and Application Development

Business Data and Management Applications

My Java desktop work includes library management with book and author registration, lease tracking, data storage and retrieval, and employee registry functionality covering employee records, departments, rankings, and task assignments. These projects strengthened my object oriented design and business data management skills while demonstrating how structured software can replace fragmented record keeping with consistent workflows.

Authentication and Web Application Development

I developed a freelance authentication interface for a United States company using HTML and JavaScript with backend integration. The work included front end design, backend connectivity, credential handling, and testing in a Linux environment. The project reinforced the importance of treating authentication as both a user experience and a security boundary.

Python Automation and Security Engineering

My Python work includes web backends, automation scripts, security testing utilities, and data processing programs. This experience strengthened my ability to build repeatable workflows and move between application development, data processing, and technical investigation.

Cybersecurity and Risk Reduction

Web Application Security Assessment

During a freelance cybersecurity engagement with InterGamma, I assessed a full stack web application using Burp Suite and OWASP ZAP with an OWASP Top 10 oriented methodology. The objective was to identify weaknesses before production release and translate technical findings into remediation priorities.

- Identified and validated more than 35 security vulnerabilities across authentication, input validation, session management, and business logic.
- Performed risk prioritization using business impact, likelihood, and severity.
- Reduced reported remediation planning time by approximately 45 percent through structured executive and technical reporting.
- Contributed to an estimated 70 percent reduction in high and critical security risks before release.
- Provided mitigation guidance covering authentication, authorization, input validation, and session security.

Business value: Security testing reduces the probability and potential cost of unauthorized access, data exposure, service disruption, and emergency remediation after deployment.

Cloud, DevOps, and Operational Reliability

My development experience includes AWS, Microsoft Azure, Google Cloud Platform, Docker, Kubernetes, GitHub, Vercel, Render, and Railway. At GWM Cleaning Services, I built and maintained web and mobile applications using the MERN stack, Python, Flutter, and Azure services. I also provided IT support, network setup, software and hardware troubleshooting, and preventive maintenance.

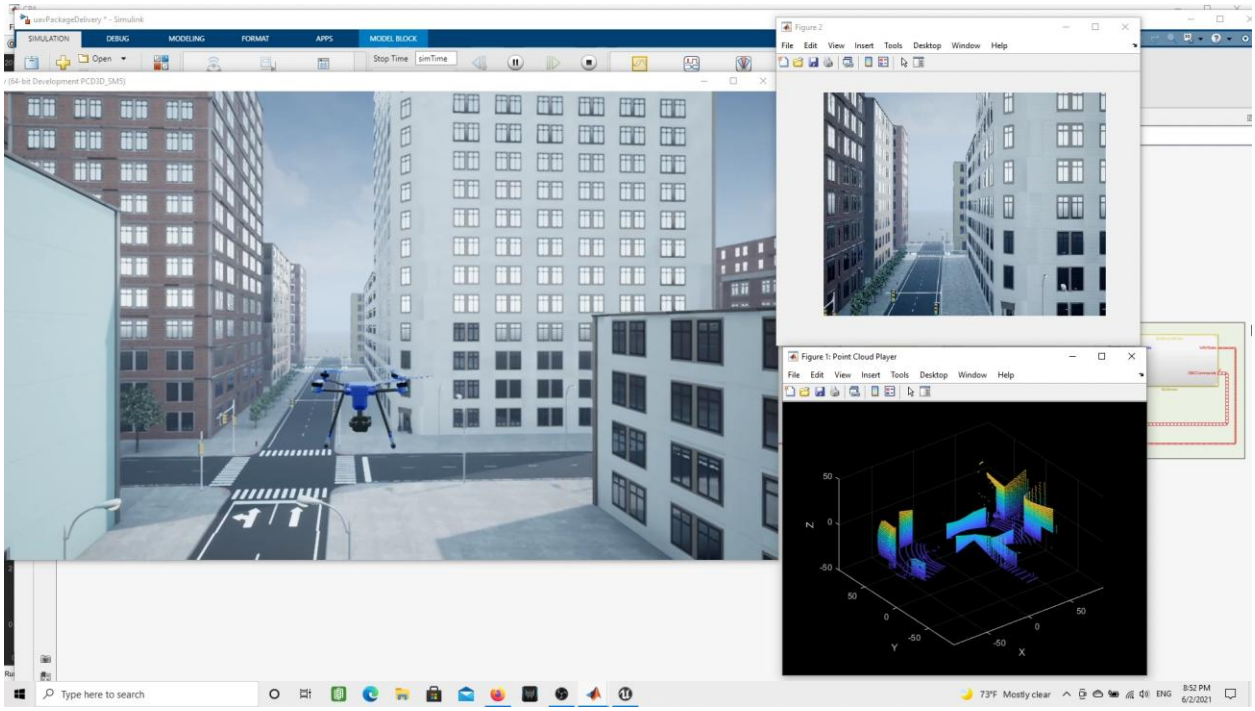
I managed inventory and maintained reported system availability of 99.9 percent through proactive monitoring and maintenance. The business value was operational continuity because reliable systems reduce interruptions for staff and customers and lower the likelihood that technical failures become business disruptions.

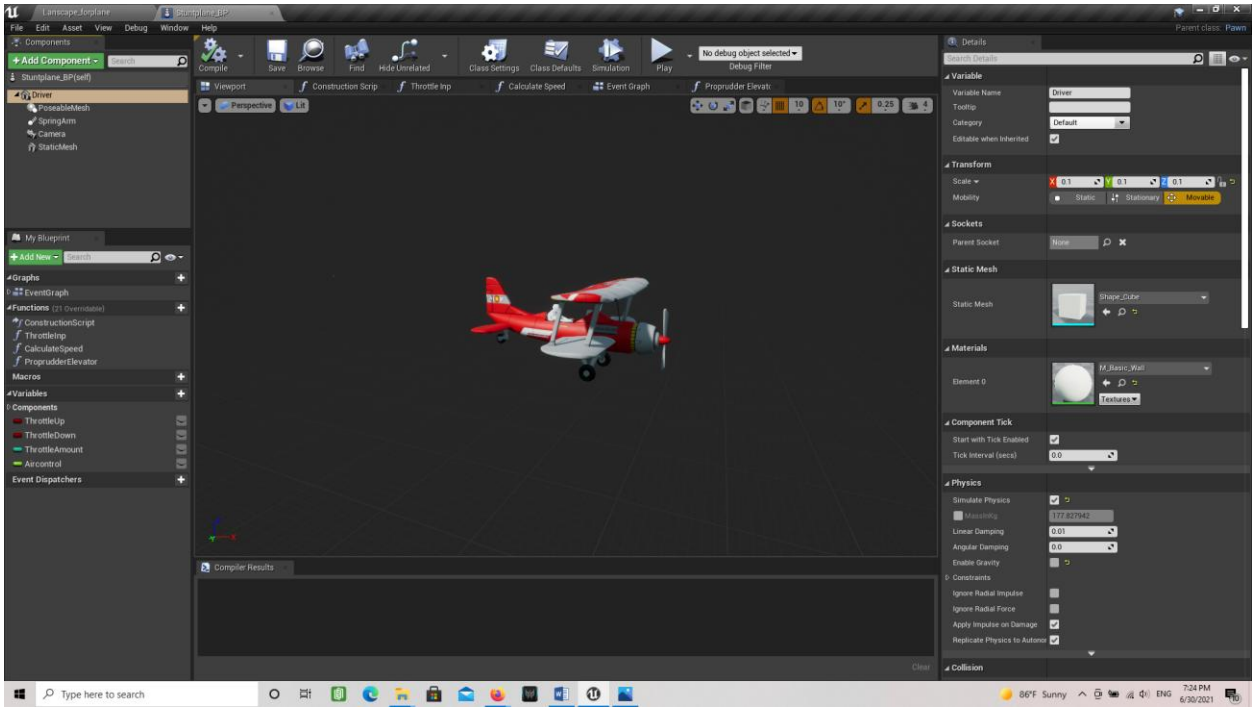
Autonomous Drone Engineering

For my Bachelor's degree in Mechatronics Engineering at the University of Balamand, I completed a final year project on autonomous drones under the supervision of Dr. Gilbert Tekli. The project combined MATLAB and Simulink with aerospace and UAV toolboxes, waypoint guidance, Lidar sensing, camera modules, GPS simulation, QGroundControl, and Unreal Engine environments.

The autonomous simulation demonstrated point to point navigation and obstacle avoidance. Additional manual control simulation included physics based movement, throttle control, torque and force, animated control surfaces, and satellite based terrain.

Business and engineering value: Simulation provides a lower cost and safer environment for validating navigation and control logic before physical deployment. It can reduce dependence on hardware during early development and expose design problems before field testing.


SIMULATION
DEBUG
MODELING
FORMAT
APPS
MODEL BLOCK
Figure 2
Figure 1: Point Cloud Player
Type here to search


Components
Add Component
My Blueprint
Viewport
Details
Compiler Results
Type here to search



Creative Engineering and Visual Design

Publication and Illustration Work

As a freelance illustrator and editor, I have worked on publication ready children's books, including original character designs, multi page layouts, atmospheric backgrounds, typography, and print ready formatting for KDP. I developed consistent character model sheets and color palettes and created high resolution front and back cover artwork.

Business value: Consistent visual systems reduce rework across a publication and create a coherent product experience. My ability to combine technical production requirements with visual design lets me contribute across engineering and creative deliverables.

Amazon book links (or just write my full name in the search bar amazon.ca/amazon.com)

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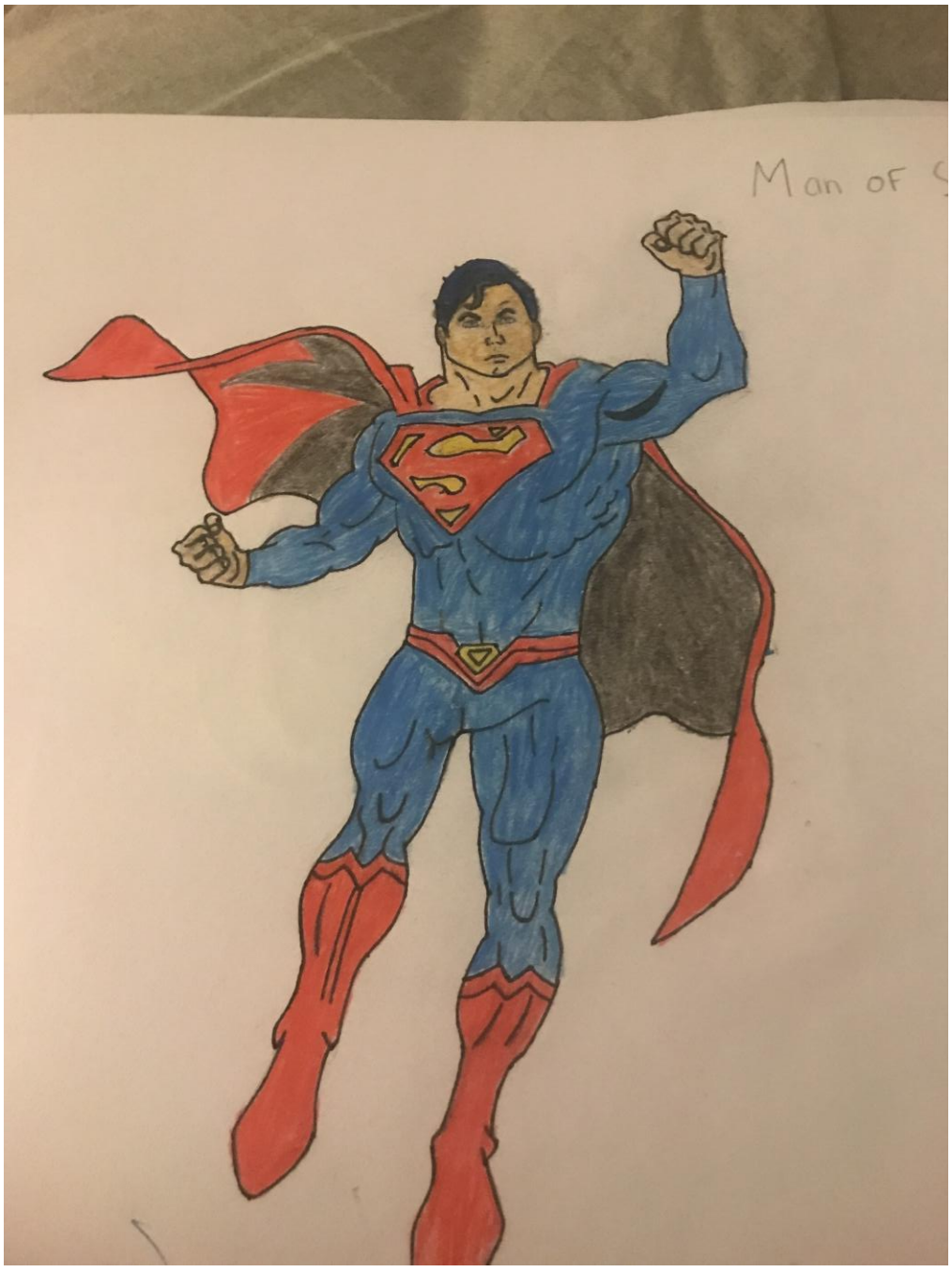
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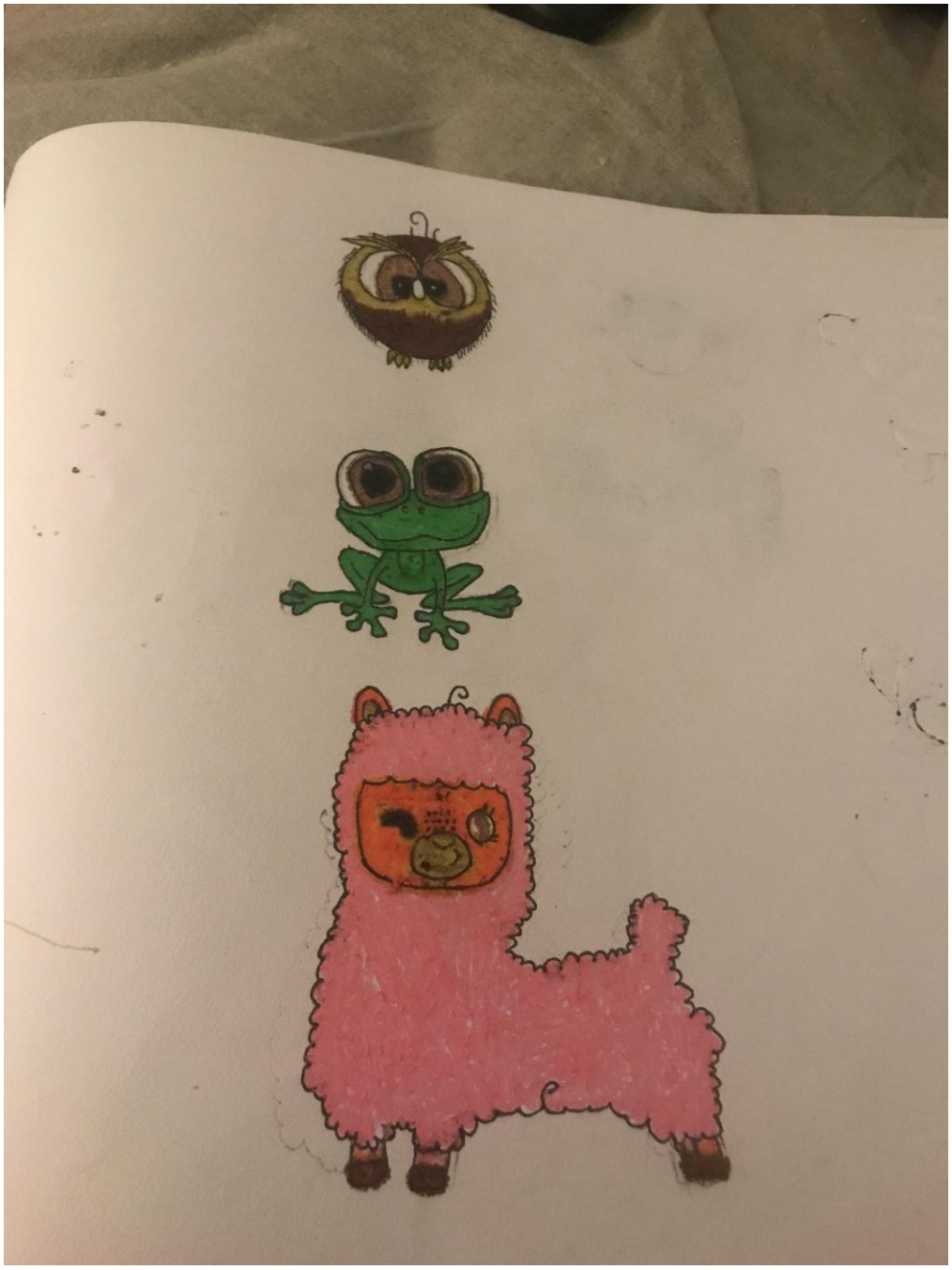
Graphic and CAD Design

My design work includes vector logo development for a United States home services company using Adobe Illustrator, Photoshop, and InDesign, along with CAD and SolidWorks work involving mechanical and technical visualization. This combination allows me to communicate ideas visually while maintaining technical precision.

Drawings







These are some of my drawings I have may more.

Technical Capabilities

- Programming: TypeScript, JavaScript, HTML, CSS, C, R, Python, Java, C++, C sharp, Dart, Kotlin, Swift.
- Frontend: Bootstrap, React, Angular, VueJS, NextJS, Tailwind CSS.
- Backend: NodeJS, ExpressJS, .NET, Spring Boot, Spring Framework, SQL, NoSQL, APIs, FastAPI.
- AI and data: Scikit Learn, Tidymodels, Caret, Keras, TensorFlow, PyTorch, Tableau, Power BI, Hadoop, Spark.

- Cloud and DevOps: GitHub, Vercel, Render, Railway, AWS, Google Cloud Platform, Microsoft Azure, Docker, Kubernetes.
- Security: Linux, web application security testing, authentication and authorization assessment, Burp Suite, OWASP ZAP.
- Design: Adobe Illustrator, Photoshop, InDesign, CAD, SolidWorks, logo design, UI design.

Education and Professional Development

Durham College, Graduate Program in Artificial Intelligence Analysis, Design and Implementation.

University of Balamand, Bachelor's Degree in Mechatronics Engineering.

Certifications and training include Cisco CCNA I, Cisco CCNA II, Cisco Introduction to Cybersecurity, Artificial Intelligence A to Z, Machine Learning A to Z, Data Science A to Z, OSINT, website hacking and penetration testing, Linux and Wireshark, bug hunting, penetration testing, and API security and API hacking.

Selected Impact Snapshot

- 30 percent reduction in data processing time reported for an LLM powered web application in the professional resume.
- 70 percent reduction in high risk security vulnerabilities reported following security assessment and remediation support.
- 38 percent reported reduction in marketing costs in Smart Leads AI project use cases.
- 31 percent reported improvement in qualified lead conversion efficiency in Smart Leads AI use cases.
- 42 percent reported reduction in acquisition and advertising costs in the Retail Intelligence pilot.
- 35 percent reported improvement in qualified lead conversion in the Retail Intelligence pilot.
- 75 percent reported reduction in manual lead qualification time in the Retail Intelligence pilot.
- 99.9 percent reported system availability managed at GWM Cleaning Services.

Professional Direction

I am seeking opportunities where software engineering, artificial intelligence, data, cloud infrastructure, and security can be used together to solve meaningful business problems. My strongest interest is in environments where technical work can be connected to measurable outcomes such as reduced processing time, improved conversion efficiency, lower operating cost, stronger security, higher availability, or better decision quality.

My approach is practical. I want to understand the business problem, build the appropriate technical solution, measure the result, and improve the system based on evidence. That mindset connects my software engineering, AI, cybersecurity, DevOps, and design experience.

Closing Statement

I bring together engineering discipline, AI and data analysis, security awareness, operational thinking, and visual communication. The common thread across my work is the search for useful outcomes. Whether the challenge is prioritizing customers, reducing security risk, maintaining reliable systems, simulating autonomous flight, or producing a publication ready design, I approach the work by asking what the system should improve and how that improvement can be measured. Thank you for taking the time to review my portfolio.